Employee Data Analysis

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INTRODUCTION

Exploring data is the first step to understand its structure, prepare it for analysis through cleaning and transformation processes, and extract useful insights from the available information.

During the data preparation stage, missing values and inconsistencies were addressed to ensure the dataset is suitable for analysis. Some variables were also transformed to improve interpretability.

Following the preparation process, exploratory analysis was conducted to examine patterns across different attributes, including salary distributions, departmental groupings, and employee characteristics.

The results of this analysis provide a clearer understanding of the dataset and highlight patterns that may support more informed organizational insights

CLEANING DATA

Data cleaning is an essential step in any data analysis process, as raw datasets often contain missing values, inconsistent formats, or unnecessary attributes that may affect the quality of the results.

Preparing Data steps :

1. check duplicates
2. check null values
3. replace null values
4. drop Exit Date column
5. create Is Exited boolean column

> The Is_Exited column was created to preserve the information from the Exit Date column. It contains boolean values indicating the employee's status: 1 if the employee has exited and 0 if the employee is still employed.

VISUALIZATION

For the visualization step, a variety of charts and plots were created to gain a deeper understanding of the dataset. This included analysing numerical features like age, salary to observe their distributions and identify trends or outliers. Categorical features such as department, job title, and Business Unit were visualized to highlight patterns in employee composition. correlation heatmaps was used to examine relationships between different variables, providing insights into factors that may influence employee behaviour. Overall, these visualizations helped reveal important trends and guided the direction of further analysis.

used plots:

01 Count Plot

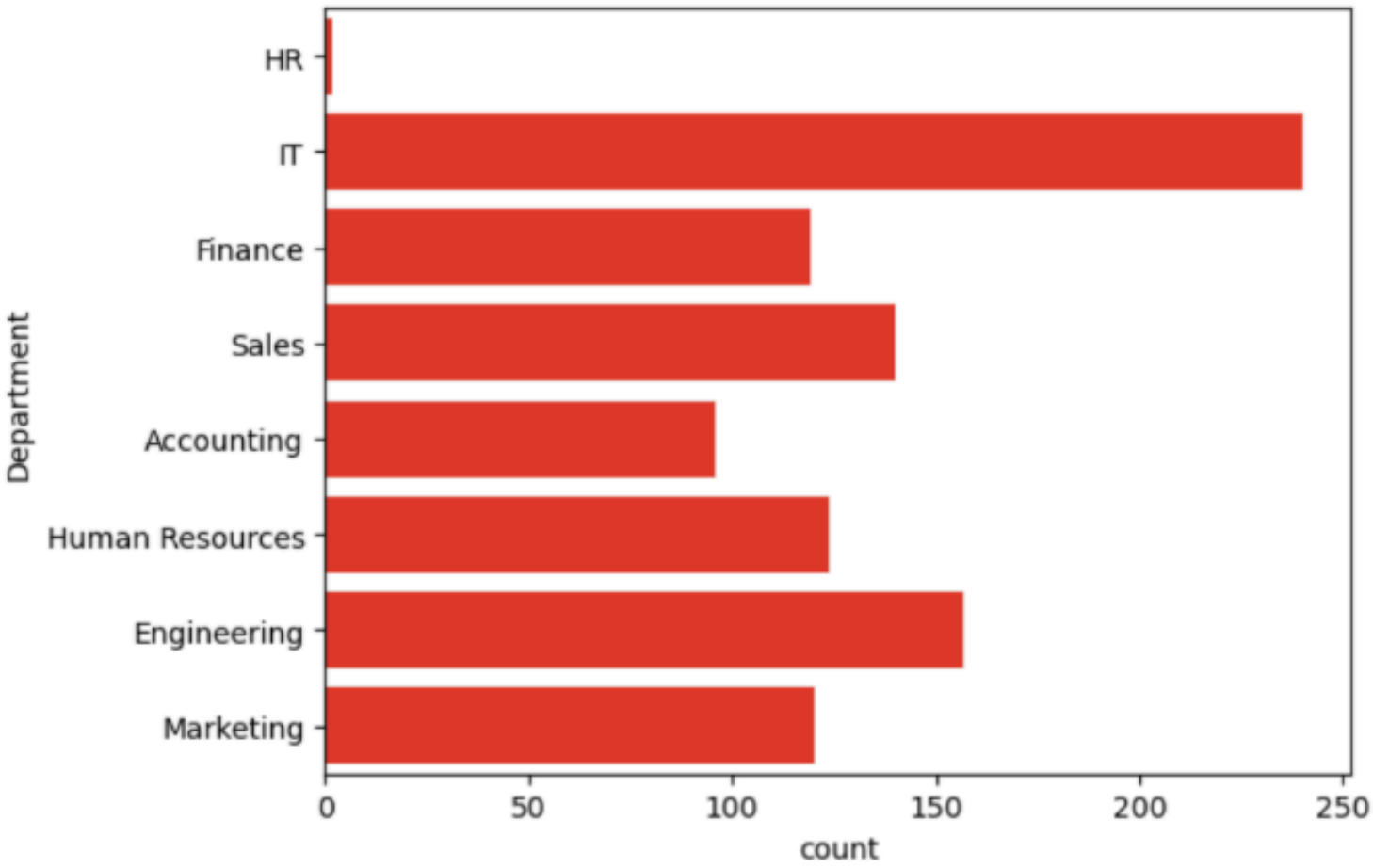
02 HeatMap

03 Scatter Plot

04 Box Plot

01- COUNT PLOT

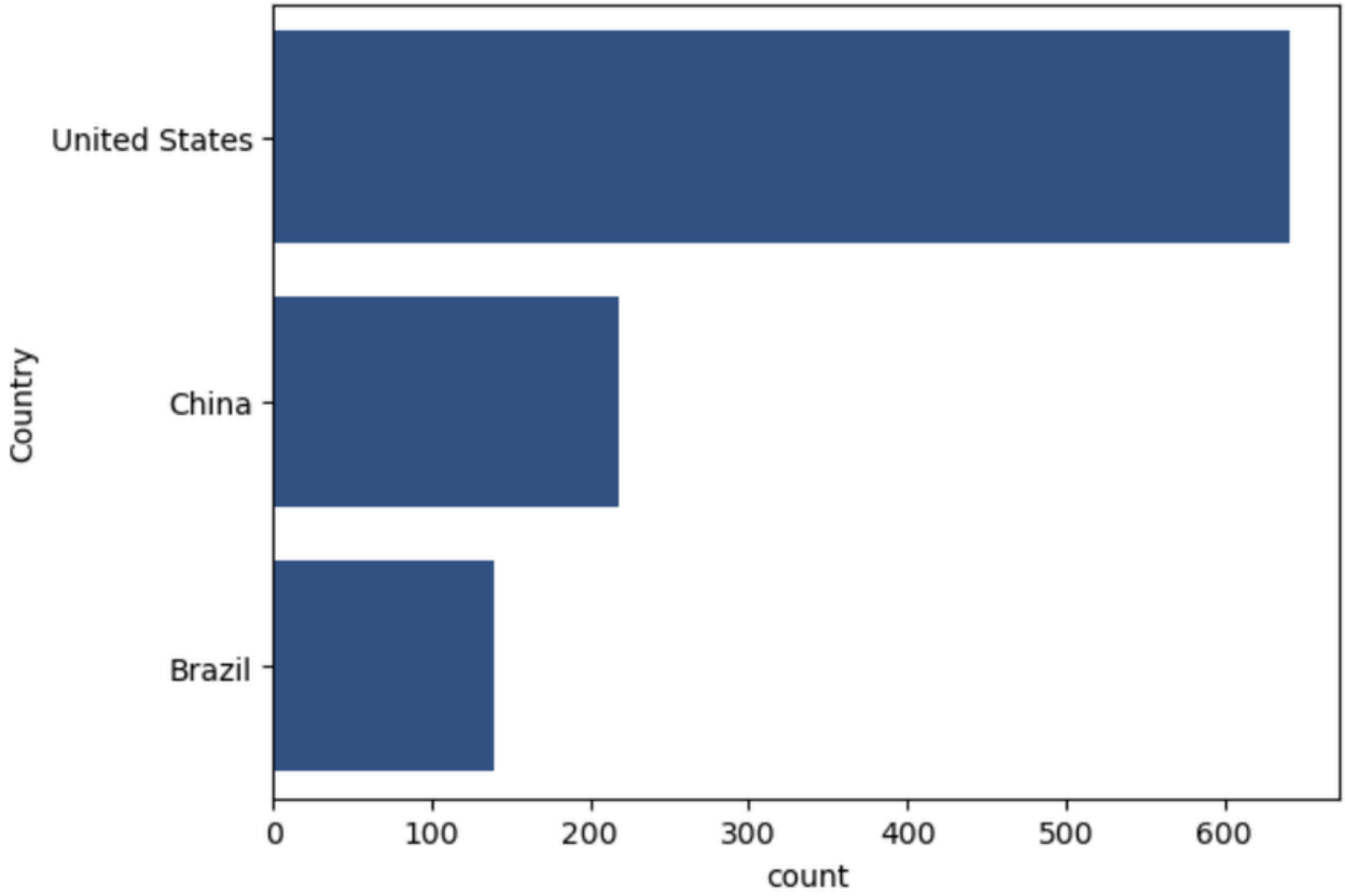
01 This plot counts the number of employees in each department



The department with the highest number of employees is IT, while HR has the fewest, which aligns with typical organizational structures.

01- COUNT PLOT

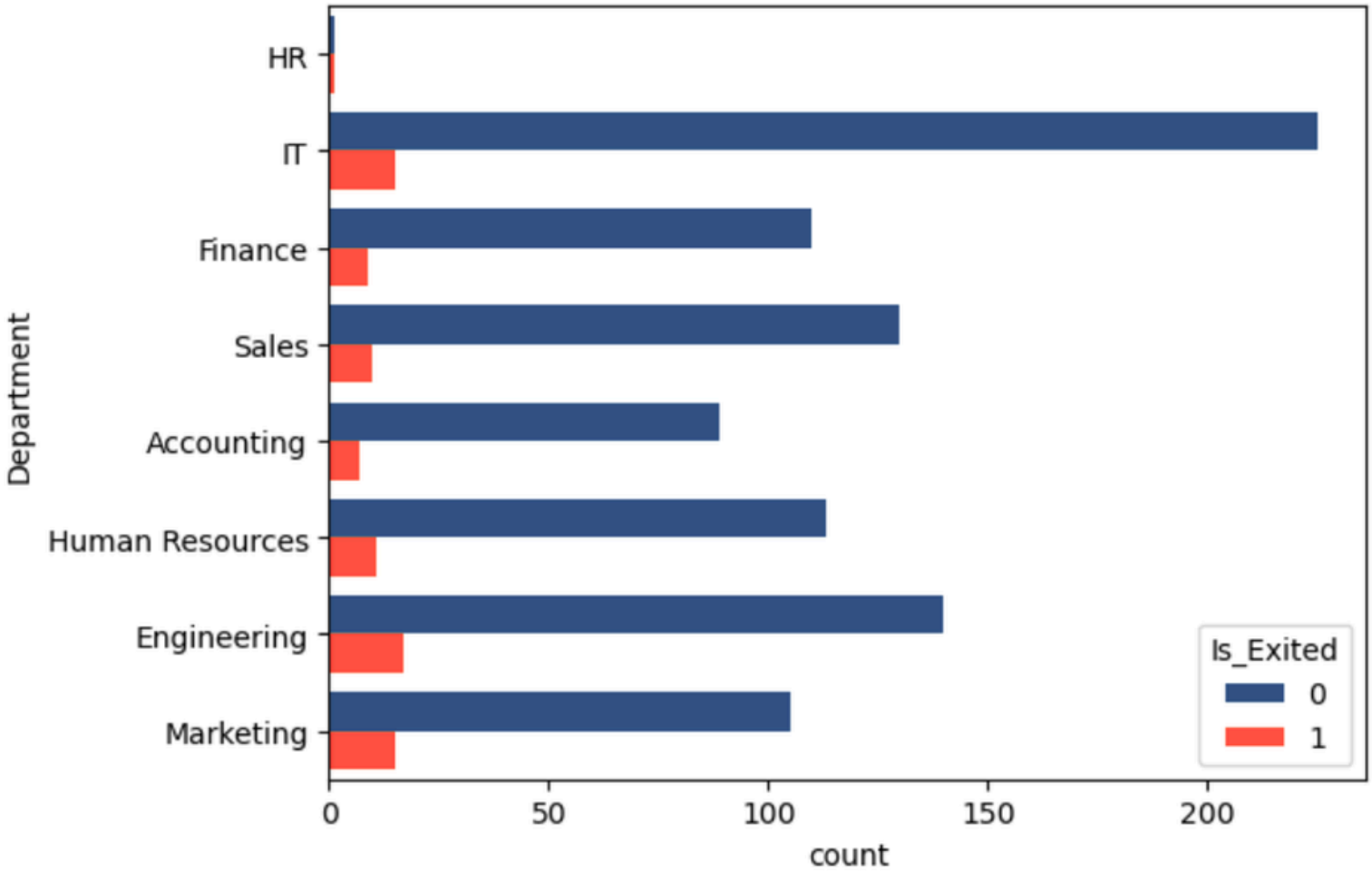
02 This plot counts the number of employees in each country



This plot shows the number of employees in each country, which helps in understanding the workforce distribution and assigning customers to the most suitable employees based on their location.

01- COUNT PLOT

03 This plot counts the number of employees in each department



This plot shows the number of employees in each department, where 0 represents employees who are still working in the company and 1 represents employees who have exited.

02- Heat Map

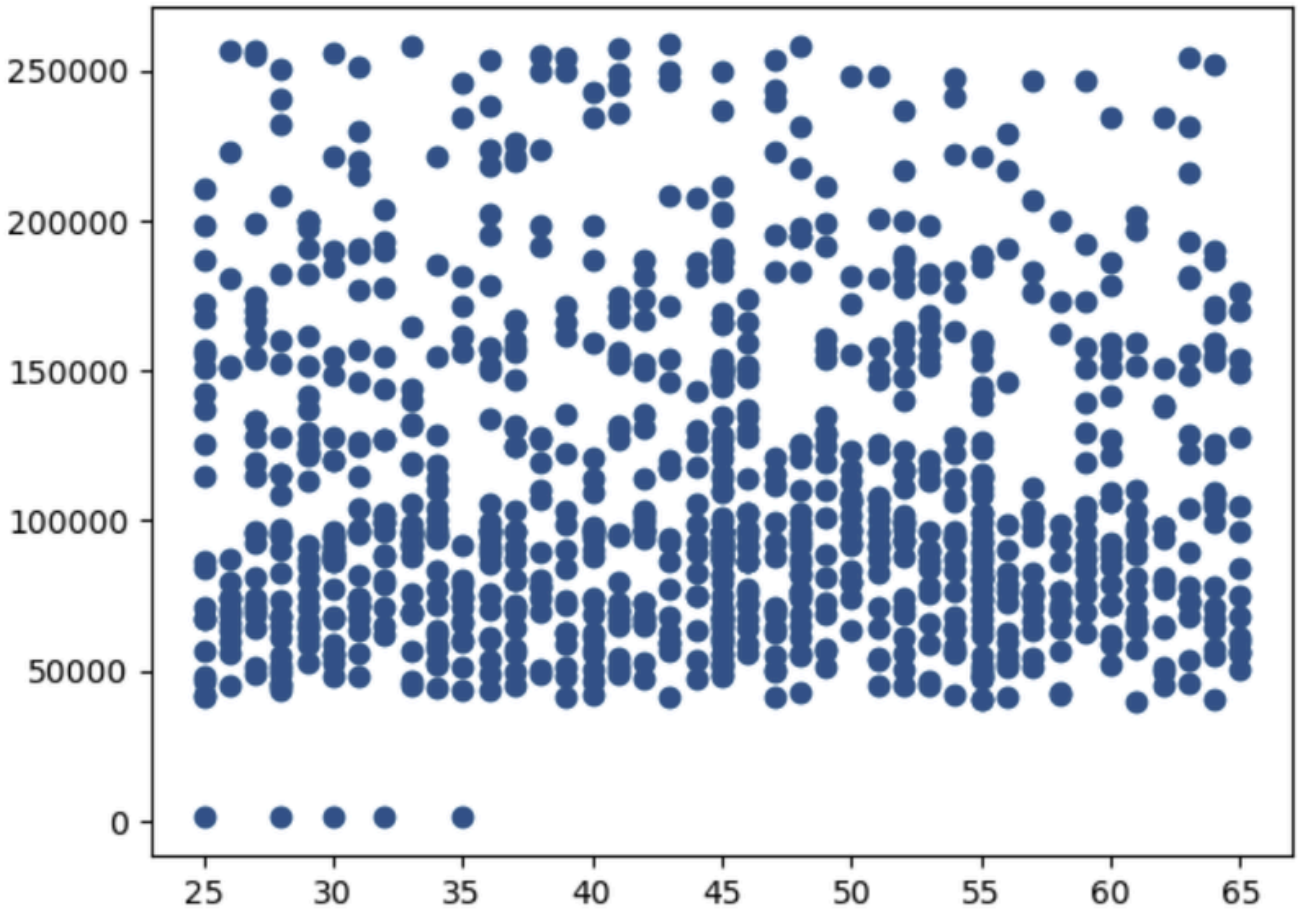
This plot represents the relationship between the Salary, Bonus and Age

The visualization shows a strong positive correlation between bonus and salary. However, their relationship with age appears weaker, indicating that employees with lower age values tend to receive relatively higher salaries and bonuses.

03- Scatter Plot

01

This plot represents the relationship between the Salary and Age

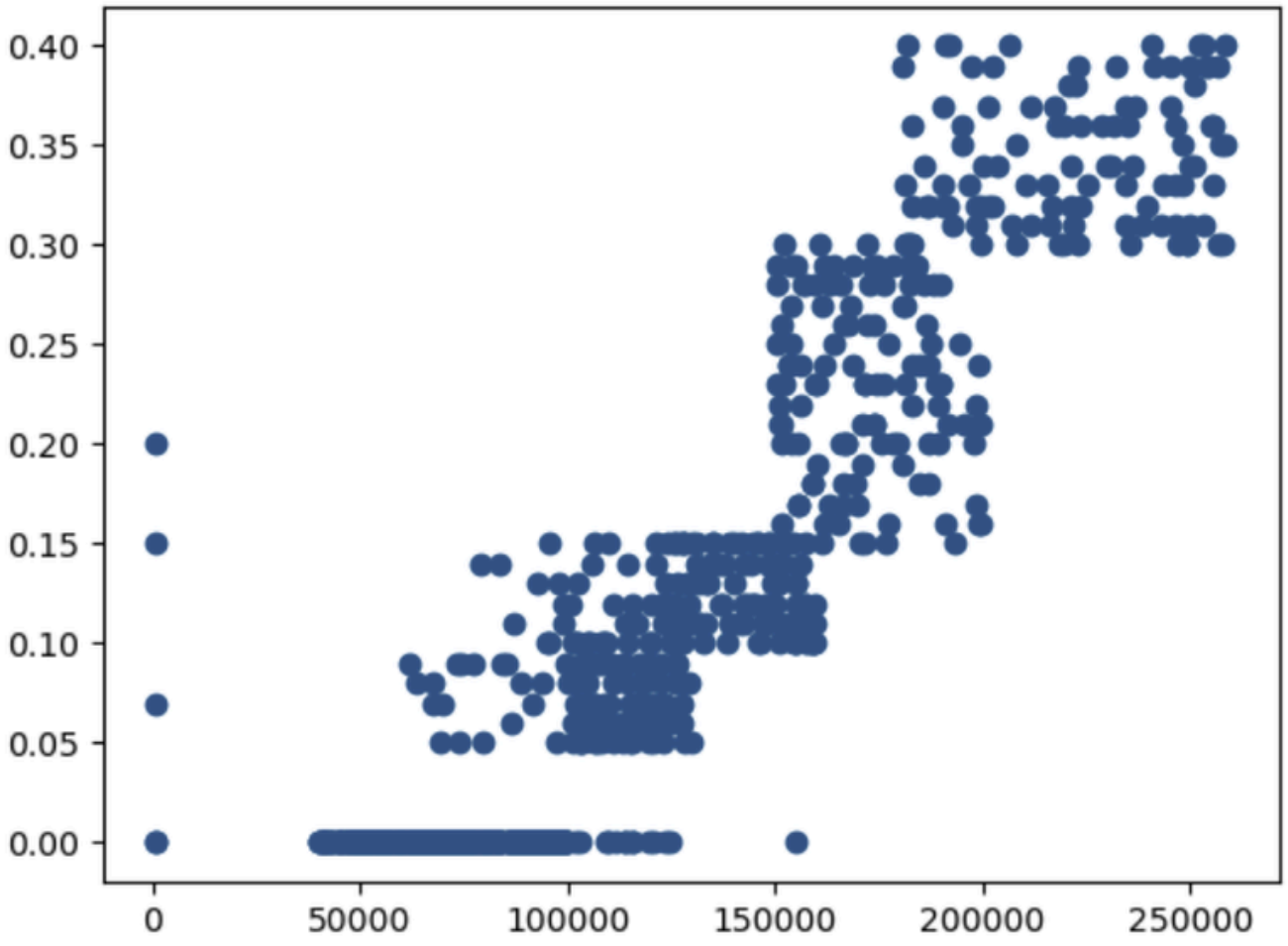


The plot shows that employees across different age groups earn salaries ranging from 50,000 to 250,000. Most of the data points are concentrated between 50,000 and 100,000, indicating that the majority of employees fall within this salary range.

03- Scatter Plot

02

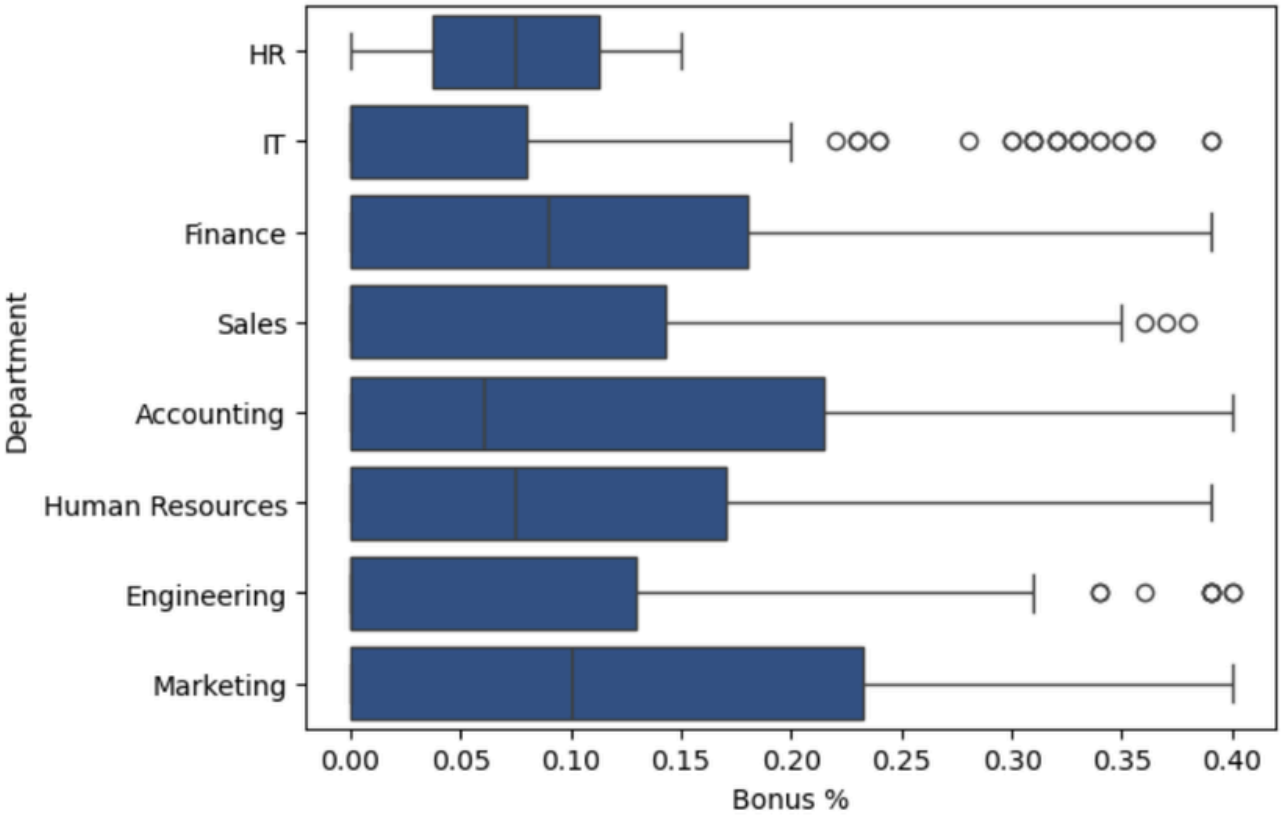
This plot represents the relationship between the Salary and Bonus



The plot of bonus versus salary reveals a step-like upward pattern, indicating that higher salaries are generally associated with higher bonuses. This pattern suggests a strong positive relationship between salary and bonus.

04- Box Plot

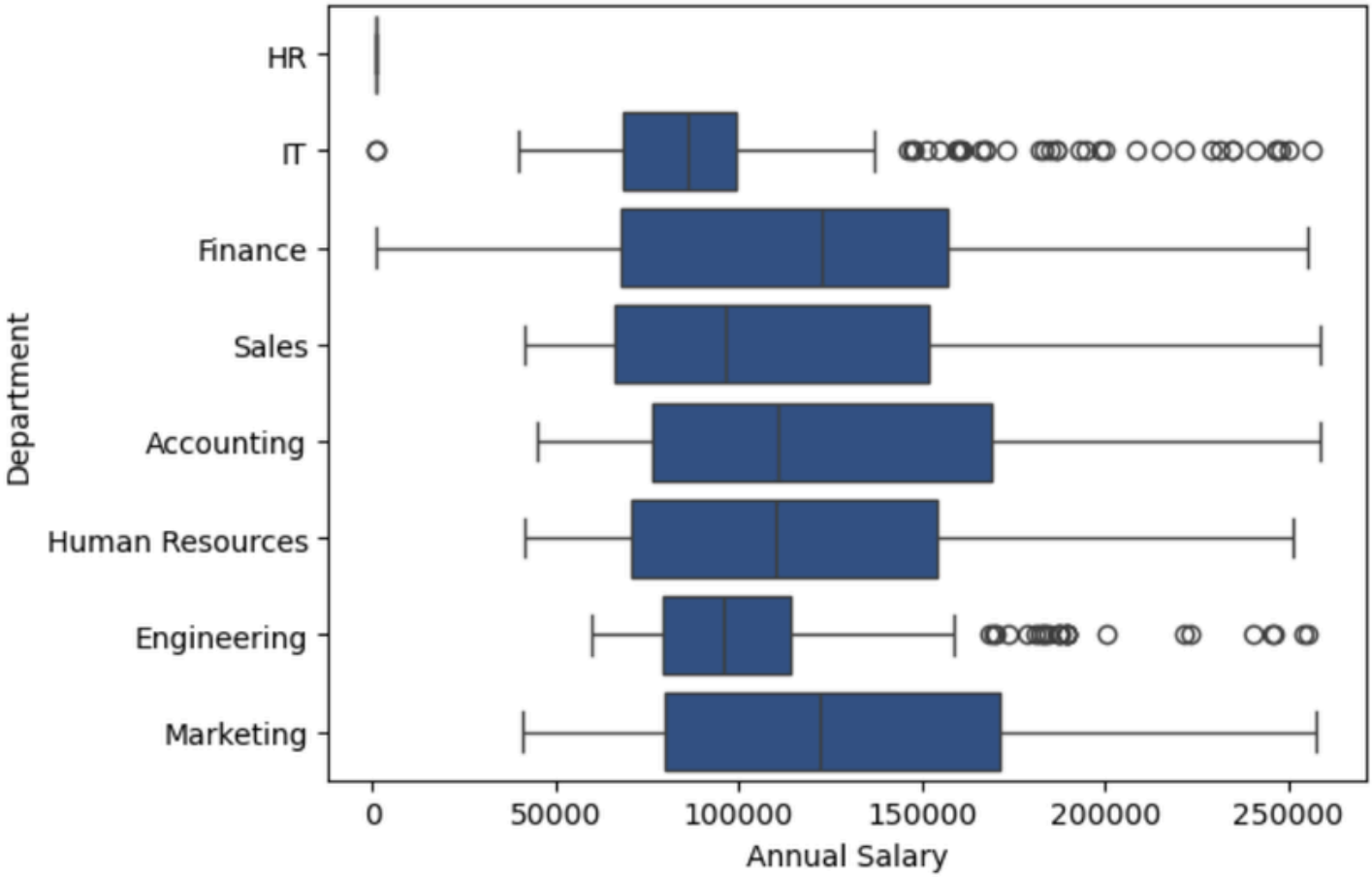
01 This plot represents the distribution between the Department and Bonus



The plot indicates that the Marketing department has the highest median value, while the IT department shows the most outliers compared to other departments.

04- Box Plot

02 This plot represents the distribution between the Department and Salary



The plot indicates that the Finance department has the highest median value, while the IT department shows the most outliers compared to other departments.

CONCLUSION

In this project, the employee dataset was analyzed to better understand workforce characteristics and identify patterns within the organization. The data was first cleaned and prepared to ensure accuracy and consistency. After that, several visualization techniques were used to explore the relationships between different features such as age, salary, bonus, department, and country.
The analysis revealed several key insights. For example, the IT department has the highest number of employees, while the HR department has the lowest.

The visualizations also showed a strong positive relationship between salary and bonus, meaning that employees with higher salaries tend to receive higher bonuses. Additionally, salary values ranged between 50,000 and 250,000, with most employees earning between 50,000 and 100,000.
Box plot analysis also highlighted differences between departments and helped identify potential outliers.

Overall, the visual exploration of the dataset provided useful insights into employee distribution, compensation patterns, and departmental differences, which can support better decision-making and further analysis.